

AMENDMENT NO. 1 APRIL 1995
TO
IS 5523 : 1983 METHODS OF TESTING ANODIC
COATINGS ON ALUMINIUM AND ITS ALLOYS

(First Revision)

(Page 4, clause 2.1) — Substitute the following for the existing clause:

‘2.1 Thickness of anodic coatings shall be measured by the microsection, stripping methods or with the help of Eddy current. The former is suitable for Grade AC 15 and thicker coatings and stripping method is suitable for Grade AC 10 and thinner coatings. The Eddy current method is suitable for any thickness from 3 to 150 microns.’

(Page 6, clause 2.3.6) — Add the following after this clause:

“2.4 Eddy Current Method

2.4.1 Eddy current test instruments which employ a probe coil carrying a radio frequency alternating current may be used. This probe is placed close to the surface of the test specimen and the electromagnetic field of the coil induces an Eddy current in the specimen by virtue of the skin effect, the Eddy current is confined to surface layer. The frequency is so chosen that the Eddy current varies with the thickness of coating, if the coating and the base metal differ appreciably in conductivity or in magnetic properties.

2.4.2 Eddy current test instruments be used to measure single layers of non-conducting coating on aluminium. Since the method depends on the conductivity of the anodic coating, it will give accurate results only if the coating to be measured has a uniform known conductivity.

2.4.3 The method of measurement of coating thickness by Eddy current shall be as given in IS 6012 : 1992 ‘Measurement of coating thickness by Eddy current method (first revision)’.”

(Page 9, clause 5.0) — Add the following ‘Note’ below this clause:

‘NOTE — In case of sulphur dioxide method and sodium sulphite immersion test, the test specimen shall be masked at the bare edges with suitable material such as stop-off lacquer.’

(*Page 14, clause 5.3.2, lines 1 to 3*) — Substitute the following for the existing:

'The surface area to be tested is degreased with a suitable solvent. A few drops of 50 percent (v/v) nitric acid is placed on the test area for 10 minutes. The test area is rinsed well and dried.'